

OMAR ABOELELLA

Computer Science & Mathematics | Quantitative Modelling | Machine Learning | Optimisation
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PROFILE

Computer Science and Mathematics student at Birkbeck, targeting First Class Honours, who builds machine learning and optimisation systems from the mathematics through to production code. Strong in Python, SQL and statistical modelling, with practical experience in Bayesian inference, predictive modelling and data pipelines. Looking for a **Sales and Trading or quantitative internship** to apply mathematical and programming skills to markets and systematic decision-making.

EDUCATION

BSc (Hons) Computer Science & Mathematics, Birkbeck, University of London Expected 2028

- **Relevant modules:** Linear Algebra, Calculus, Discrete Mathematics, Data Modelling and Analysis, Systems Analysis.
- Target: First Class Honours.

Also: Access to HE Engineering (Distinction), Westminster Kingsway 2025; IBM Data Science; Google Mathematics for ML; Microsoft Fabric; 7 GCSEs including Maths (A).

TECHNICAL SKILLS

Programming Python (NumPy, Pandas, SciPy, FastAPI), SQL, PostgreSQL, LaTeX, JavaScript, HTML, CSS

ML & Statistics Predictive modelling, Bayesian inference, time-series forecasting, feature engineering, RAG, MAPE-K reasoning loops

Quantitative Linear algebra, calculus, optimisation, probability and statistics, data modelling and analysis

Engineering Git, GitHub, Docker, CI/CD, containerisation, data pipelines, automation, Jira, Excel

Languages Arabic (native), English (fluent)

TECHNICAL EXPERIENCE

AI Product Engineer, Digis Squared Ltd Jan 2026 - Present

- Set the technical strategy for the KATANA autonomy programme with the product manager, moving the platform from Level 1 to Level 4 against TM Forum standards.
- Designed the closed-loop MAPE-K reasoning architecture using Bayesian inference to automate network decisions.
- Built total cost of ownership (TCO) models and ran security auditing across the platform's sub-modules.
- Planned sprints for the development and test teams and created cross-team workflows in LangGraph, including how data is selected and accessed for the data team.

AI Product Intern, Platform & Ops, Bitget Feb 2026 - May 2026

- Analysed exchange telemetry and built predictive models on historical and live market data to forecast price movement.

R&D Engineering Trainee, Digis Squared Ltd Sep 2024 - Mar 2025

- Researched network systems, focusing on hardware and software integration.

RESEARCH & TEACHING

Fintech Research Placement, Optimisation & ML, Santander Incoming, Summer 2027

- Selected for a research placement applying mathematical optimisation and machine learning to **credit-scoring models**.

Independent Research, AI in Higher Education 2026 - Present

- Investigating how efficiently AI models complete university assignments and the effect on student learning outcomes.

Mathematics & Physics Teaching Assistant, Southfields Academy Mar 2026 - Present

- Teach GCSE Higher maths and physics to international and EAL students, explaining quantitative concepts clearly under time pressure.

Market Intelligence Research Intern, BP Jun 2024 - Sep 2024

- Localised models for the offshore-wind market intelligence team and automated the department's competitor-research reporting.

SELECTED PROJECTS

- **Crypto Prediction Models.** Machine learning models forecasting Bitcoin price movement from historical and live market data. (Bitget)
- **Competitor Intelligence Engine.** Automated weekly pipeline that pulls competitor data from Bloomberg and other sources, analyses it and produces Excel reports for the market intelligence team. (BP)
- **OctiMind and INOS.** Network testing and optimisation products deployed to O2. (Digis Squared)
- **NOOKBASE, Founder.** Agentic study platform for iOS, Android and web with a conversational tutoring agent. Around 150 beta users, pre-launch.
- **Quantitative self-study (ongoing).** Self-directed work across optimisation, probability and stochastic processes, and quantum computing with Qiskit.